# MICROCONTROLLER COURSE PROJECT

# Project Idea Proposal

Course: Microcontroller Laboratory / Project

Department: Computer Science & Engineering (Cyber Security)

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Date: ___________________

## Executive Summary

This document presents five shortlisted embedded systems project concepts identified through a structured technology scouting, opportunity analysis, portfolio optimization, and concept synthesis process. The ideas were selected based on innovation potential, technical feasibility within a two-month semester timeline, affordability (target prototype budget of ₹3,000–₹5,000), demonstration quality, commercialization potential, and scope for future patent investigation.

## Idea 1 – Affordable Refreshable Braille Display

We are proposing the development of a low-cost refreshable Braille display aimed at improving accessibility for visually impaired users. Existing refreshable Braille devices remain prohibitively expensive due to complex actuation mechanisms. Our proposed system investigates an alternative embedded approach capable of dynamically generating tactile Braille characters while reducing overall system cost and complexity. The objective is to make digital reading and communication more accessible and provide a practical foundation for future assistive technologies.

## Idea 2 – Intelligent Bearing Fault Early-Warning System

We are proposing an intelligent predictive maintenance system that continuously monitors vibration patterns in rotating machinery to detect early signs of bearing degradation. Instead of relying solely on scheduled maintenance or manual inspections, the proposed solution aims to identify faults before catastrophic failure occurs. This approach can improve equipment reliability, reduce maintenance costs, and demonstrate how embedded intelligence can support modern industrial automation.

## Idea 3 – Self-Correcting Smart Soil Monitoring System

We are proposing an intelligent soil monitoring system capable of identifying measurement drift and maintaining reliable soil moisture estimation over extended periods. Conventional low-cost sensors gradually lose accuracy due to environmental conditions and aging. Our proposed system aims to improve measurement reliability through adaptive monitoring and self-correction, helping reduce water consumption while supporting precision agriculture.

## Idea 4 – Embedded Firmware Integrity and Secure Boot Verification

We are proposing an embedded security system that verifies firmware authenticity before device startup to prevent unauthorized or tampered software from executing. As connected embedded devices become increasingly common, firmware integrity has become a critical cybersecurity challenge. The proposed solution focuses on strengthening device trust, improving resilience against firmware attacks, and demonstrating practical embedded cybersecurity principles for future IoT and industrial systems.

## Idea 5 – Intelligent Cold Chain Integrity Monitoring System

We are proposing an intelligent monitoring system that continuously evaluates environmental conditions affecting temperature-sensitive products during transportation and storage. Rather than relying on temperature alone, the proposed approach considers multiple conditions to provide a more comprehensive assessment of cold chain integrity. The objective is to improve healthcare logistics, reduce product wastage, and enhance quality assurance throughout the supply chain.